

Preface

This manual provides the information needed to operate the WRO 300.

Valid program version	3.x
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Definitions of expressions in the manual



WARNING

Is used to alert the user/operator **not to take** a certain action, which if taken can cause a potential hazard and result in a serious adverse reaction, injury or death. A warning may also be used to alert the user/operator **to take** a certain action to avoid the potential hazard as above.



CAUTION

Is used to alert the user/operator **to take** a certain action to protect against a potential hazard which, if ignored, could have an adverse effect on the patient or the device. A caution may also be used to alert the user/operator **not to take** a certain action to avoid the potential hazard as above.

NOTE

A reminder to the user/operator on normal treatment activity and on what is a suitable action in a particular situation.

Safety considerations



WARNING

Unauthorized installation, modifications, alterations or repair of the WRO 300 may result in malfunctioning or have other serious consequences for the safe operation of the equipment.



CAUTION

1. WRO 300 may be operated only by persons who have studied the instructions in this manual and the manual for the dialysis monitor. If the system does not perform as described in this manuals, it should not be used until the condition is rectified.
2. The operator should pay attention to alarms and follow the instructions, warnings, cautions and notes given in the manual.
3. The WRO 300 will perform as designed only if it is used and maintained in accordance with Gambro's written instructions. Any warranties made by Gambro with respect to the WRO 300 are voided if the equipment is not used in accordance with the written instructions provided. Gambro will not accept responsibility for any damage or injury resulting from improper use or maintenance or unauthorized repair.
4. The use of mobile telephones or communication equipment in the vicinity of the WRO 300 could adversely influence the performance of the machine. See specification.
5. WRO 300 needs special precautions regarding EMC and needs to be installed and put into service according to the EMC information provided in the operators and service manual.
6. For USA: US Federal law restricts this device to sale by or on the order of a physician.

NOTE

1. During transportation and storage the equipment must be kept in its original packing. If transportation or storage time is more than 15 weeks the environmental data relating to the operation must be followed.
2. For the qualified technician the WRO 300 Service Manual is available. The Service Manual provides all of the necessary information for the safe and required maintenance of the machine.

CE-marking

This product is CE marked in accordance with the requirements in EC Council Directive 93/42/EEC of 14 June 1993 concerning medical devices.

Notified body is: British Standards Institution (BSI) with the notified body number 0086.

The symbol below means: Year of manufacturing.



The CE-marking of this manual is only valid if the device which it describes is CE-marked.

EC-Declaration of Conformity

We

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being the manufacturer within the European Union of the following equipment:

WRO 300
Part No. W100001001

declare that the above mentioned device is in conformity with the relevant provisions of the Council Directive 93/42/EEC of 14 June 1993 concerning medical devices, Annex I -Essential Requirements-, and our company has been subjected to the procedures laid down in Annex II -full quality assurance system- of the above mentioned directive under the supervision of the British Standards Institution, a Notified Body authorized by the United Kingdom Competent Authority, and carrying the Notified Body Number 0086.

This device does not incorporate, as an integral part, a substance as referred to in above mentioned Directive Annex I Section 7.4.

Lund, Sweden
2004-09-27


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1.Introduction

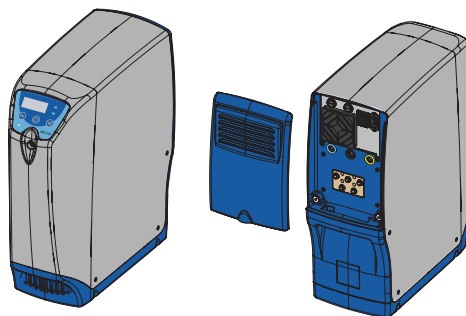
1.1 Intended use

The Gambro WRO 300 water purification unit is designed to be used as a dialysis accessory device to produce water used to prepare and dilute dialysis concentrate to form dialysis fluid by using the reverse osmosis concept. It is intended for use in conjunction with one dialysis machine, provided that the input flow and pressure demands correspond to the output of the WRO 300 unit.



CAUTION

The water produced by WRO 300 should be analysed on a regular basis to verify that it conforms with applicable regulations or standards for water for dialysis.



1.2 General

The WRO 300 has been designed to fulfil the special requirements of dialysis setting. For this purpose this device has the following features:

- Simple user interface
- Automated Chemical disinfection and cleaning modes
- Flush to drain function during Standby periods to minimize stagnant water.

This manual for WRO 300 includes instructions for operating, monitoring, cleaning, disinfecting and troubleshooting this device from an operator's perspective. For installation instructions refer to the enclosed *Installation Guide*.

1.2.1 Conductivity monitoring

The conductivity monitoring system in the WRO 300 has three limits: notification, alarm and stop. To change the settings of the limits, refer to the *Service Manual*.

NOTE

If the notification, alarm and stop limits are set to the same value the priority of the action are in the order of stop, alarm and notification.

Conductivity notification

The value of this notification is preset to 60 $\mu\text{S}/\text{cm}$ at delivery. If set (by an qualified technician) to a value slightly above the normal product water conductivity, this limit will inform about a slight increase compared to "normal" values, which may indicate incipient fouling, degradation of the RO-membrane or change in the feed water quality. The conductivity notification will appear when WRO 300 goes to Standby if the set value has been exceeded for more than 30 seconds.

NOTE

If the feed water conductivity is known to vary significantly, the conductivity notification limit should be inactivated by setting it at the same value as the conductivity alarm, see below.

Conductivity alarm

The value of this alarm limit is preset to 60 $\mu\text{S}/\text{cm}$ at delivery. If, during normal operation, the conductivity of the product water exceeds this alarm limit for more than 30 seconds, an audible and visual alarm will be issued. The WRO 300 will, however, continue in operation. The buzzer will be muted indefinitely by pressing the



button, however the button remains lit to indicate that the alarm persists.

Conductivity stop

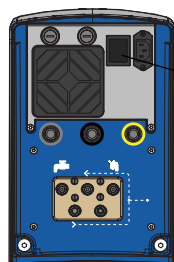
The value of this stop limit is preset to 60 $\mu\text{S}/\text{cm}$ at delivery. If, during normal operation, the conductivity of the product water exceeds this stop value for more than 30 seconds, an audible and visual alarm will be issued and WRO 300 will stop. The alarm can be acknowledged by pressing . The WRO 300 can be started again but the alarm will reappear after 30 seconds if the limit still is exceeded.

Low rejection rate notification

The value of this notification is preset at 90% at delivery. It will be issued when WRO 300 goes to Standby next time if the set notification value is exceeded for more than 5 minutes. It will inform the operator that there is a decrease in the membrane performance compared to "normal" values, which may indicate incipient fouling or degradation of the RO-membrane.

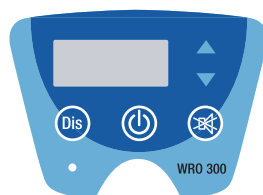
1.2.2 Mains power switch

The mains power switch is only used in service situations to turn off the power. The mains power switch is located on the rear side behind a detachable cover.



Mains power switch

1.3 Operator's panel



1.3.1 Description of buttons

Buttons	Description
	Press for three seconds to start operation (RUN).
	Press to stop operation (STANDBY).
	Press for three seconds to initiate Chemical disinfection. Only valid when the disinfection wand is connected.
	Press for three seconds during the chemical intake procedure to pause.
	Press for three seconds during dwell period to initiate rinse.
	Flashes in alarm situations. Press to silence audible alarms and to get alarm info text.
	Used to scroll and view operational data shown in the display. Pressing both arrows simultaneously will lock current information in the display. Press any arrow key to release the locking function.

1.3.2 Description of indications

Indications			Description
	Green	General	Indicates that the WRO 300 is energized. The mains power switch is located behind detachable cover at rear.
	Green	Steady light	RUN mode.
		Slow flash	The flush to drain function during Standby is activated.
		Fast flash	Insufficient feed water supply during operation.
	Yellow	Steady light	Chemical disinfection or Cleaning is in progress.
		Medium flash	The disinfection wand connector is inserted into the chemical intake.
			The chemical cycle is paused. Forced rinse is required (404 RINSE REQUIRED).
	Red	Medium flash	Notification, alarm or stop condition has occurred.
		Steady light	Notification, alarm or stop condition has been confirmed. To view the alarm list press the arrow buttons.

Definition of flash indications

Flash indication	Time	
	On	Off
Slow flash	0.4 sec	2.0 sec
Medium flash	0.4 sec	0.4 sec
Fast flash	0.2 sec	0.2 sec

1.3.3 Reminders

Notifications can be preset to remind the user do certain actions, for example exchange cartridge in the pre-filter. Further information of setting the Reminders, refer to *Service Manual*.

1.3.4 Operational data

The arrow buttons are used to scroll and view operational data in the display.

Operational data		Unit
PRODUCT WATER		X μS/cm
FEED WATER		X μS/cm
REJECTION RATE		X %
PRODUCT WATER TEMP		X °C
CHEM,	DAYS SINCE LAST ^a	X
ACID CLEAN,		X
ALKALINE CLEAN,		X
PROGRAM VERSION		X.y
TOTAL RUN TIME		X hours
DATE		yyyy-mm-dd
TIME		hh:mm
INTERNAL SERVICE		N/A

a. Visible depending on settings

In alarm situations the arrow buttons can be used to scroll through the alarm list. Refer to *“Troubleshooting” on page 5:1*.

2.Operation

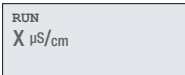


WARNING

This device does not remove chlorine and chloramines. If these substances are present in the feed water, carbon filtration is required to remove these substances. Severe patient injury may otherwise occur. A test for total chlorine of the pretreated water must be performed prior to initiating dialysis treatment. The level of total chlorine must be below 0.1 mg/l (ppm).

2.1 Start

Press  for three seconds (light on).



If Remote start/stop is provided (Not available in the USA), WRO 300 will start automatically when requested from the dialysis machine.

2.2 Stop

Press  (light off) and the WRO 300 goes to Standby mode.



If remote start/stop is provided (Not available in the USA), the WRO 300 will stop only when the dialysis machine is switched off.

NOTE

If power failure occurs during operation mode the procedure continues in the same phase as when the power failure occurred when the WRO 300 starts up again.

3.Regular maintenance

3.1 Chemical disinfection

To perform any of the following procedures, each protocol must be defined before performing the procedures in these chapter.

The required Chemical disinfection frequency to fulfil the desired microbiological requirements for the product water depends on several factors, such as:

- the quality of the feed water
- local regulations regarding the microbiological quality of dialysis water, culturing techniques etc.
- local preferences, for example regarding use of Chemicals

Therefore, no general rules can be given that cover all situations. The Chemical disinfection schedule should instead be based upon microbiological testing performed by the clinic. A maximum period between disinfections should then be established to ensure acceptable bacteriological quality of the product water according to the clinic's standards.

As a guideline, Gambro recommends a minimum frequency of weekly Chemical disinfection to ensure consistent microbiological quality of the product water.

The use of an ultrafilter on the dialysis machine will remove possible contaminants originating from the product water and may therefore be used as a method to extend the time between Chemical disinfection of the WRO 300.

If the WRO 300 will not be used for an extended period of time, weekly Chemical disinfection in combination with the built-in flush to drain function during Standby will help maintain the microbiological quality of the product water.

For certain peracetic acid disinfectant the manufacturer recommends that the WRO 300 should be cleaned with an acid cleaning agent per the cleaning instructions prior to Chemical disinfection to help preserve the RO membrane. Refer to *"Cleaning" on page 3:8*.

Contact Gambro for suitable chemical agents for the procedures.

3.1.1 Approved chemical disinfectants

- Dialox, undiluted (trade mark of Seppic, France).
Required amount: 130 ml per disinfection cycle (will be diluted to 3 % solution of Dialox by the WRO 300).
- Minncare® Cold Sterilant or Renalin 100
Required amount: 50 ml of concentrated disinfectant
- Other peracetic acid disinfectants, provided that they are approved by the manufacturer for disinfection of thin film composite reverse osmosis membranes made of modified polyamide. Follow the *manufacturer's instructions* for Use for the specific disinfectant.
The internal fluid volume of WRO 300 is approximately 4.5 litres.

CAUTION

Disinfectants may be toxic. Refer to *manufacturer's instructions*.

CAUTION

A test for residual disinfectant after rinse must be performed before the initiation of the next dialysis session. The residual concentration of the disinfectant in the fluid must be below levels specified by the clinic or by national standards. It is essential to use an appropriate test method, either with proven sensitivity for the disinfectant or recommended by the manufacturer of the disinfectant.

Example of test kit

SteriChek™ Residual Peroxide, Gambro order no. 103428.

3.1.2 Chemical disinfection procedure

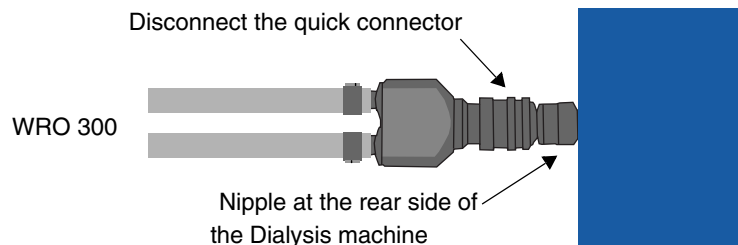
NOTE

If power failure occurs during Chemical disinfection **404 RINSE REQUIRED** is displayed when the WRO 300 restarts. Press  for three seconds to initiate rinse.

- 1 Disconnect the WRO 300 from the dialysis machine with the quick connector on the product water loop. Alternatively the connector can be left in place. However note the caution below.

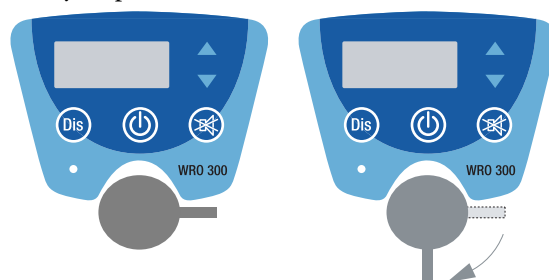
CAUTION

If the WRO 300 is not disconnected from the dialysis machine during the Chemical disinfection procedure, the test for residuals after the Chemical disinfection procedure must be performed on the dialysis machine according to the *operator's manual* of this device.



- 2 Place a tag on the WRO 300, stating that Chemical disinfection is in progress and what type of disinfectant is being used.

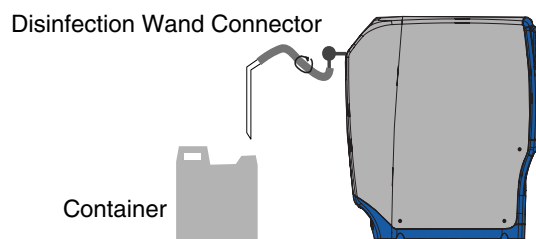
- 3 Insert the wand connector into the chemical intake port of the WRO 300 (see figure below), press firmly and turn the connector downwards to lock. Check that the connector is securely in place.



Insert the connector and press

Turn to lock

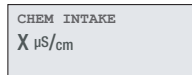
- 4 **CHEM SELECT** is shown in the display and **Dis** starts to flash.
- 5 Insert the open end of the disinfection wand into a container with sufficient amount of disinfectant (refer to [“Approved chemical disinfectants” on page 3:2](#)). Check that the open end of the intake wand is below the solution level and that the clamp is open.



- 6 From **CHEM SELECT**, press down arrow button  to enter the list of available Cleaning and Chemical disinfection programs.
- 7 Select preferred Chemical disinfection program by using the arrow buttons.



- 8 Initiate the disinfectant intake by pressing  for three seconds (steady light).



NOTE

The display shows present phase of the Chemical disinfection procedure.

NOTE

The container can be empty before the intake phase is finished.

Pause, interrupt or restart procedure

	Action
Pause of chemical intake phase	Press  for three seconds. To continue the intake press  for three seconds again.
Interrupt Chemical disinfection procedure	Disconnect the wand connector and press  for three seconds to initiate rinse.
Restart of intake phase	Remove the wand connector and insert it again. Then press  for three seconds.

- 9 When the buzzer sounds and  is flashing, press . **401 REMOVE WAND** is displayed.



- 10 Keep the wand in the container and disconnect the wand by turning the connector a quarter of a turn and gently pull it out. Let the disinfectant liquid from the line and wand flow back into the container. Then clamp the line. Also check that correct amount of disinfectant has been consumed.
- 11 Remove the disinfection wand from the container and flush with water.

- 12 The Chemical disinfection procedure will now continue automatically with *DWELL PERIOD* followed by *RINSE*. The remaining time to complete the Chemical disinfection procedure is also displayed.

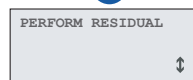


NOTE

The conductivity value is checked during the final five minutes of the rinse phase. If the value is above the Conductivity notification limit (refer to [“Conductivity monitoring” on page 1:2](#)) when the preset rinse time is finished it is prolonged 30 minutes. If the value decreases below the Conductivity notification limit the prolonged rinse finishes.

If the level still is above the Conductivity notification limit when the prolonged rinse is finished the alarm **403 HIGH CONDUCTIVITY RINSE** appears.

- 13 When the Chemical disinfection program is completed the display will show *PERFORM RESIDUAL TEST*. The indication  will remain steady lit and the  will flash.



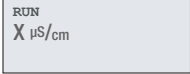
3.1.3 Residual test after Chemical disinfection



CAUTION

If the WRO 300 has been connected to the dialysis machine during the Chemical disinfection procedure, the test for residuals after the Chemical disinfection procedure must be performed on the dialysis machine according to the *operator's manual of the dialysis machine*.

- 1 Press  for three seconds to start WRO 300. Indication  will be steady lit and  will flash.
- 2 Collect a water sample either from the product water line or from the dialysis machine according to the caution above.
- 3 Test for disinfectant residuals with Sterichek Residual Peroxide test strips or other approved test strips or methods.
- 4 Confirm by pressing  until light goes out (three seconds). Indication  remains steady lit



RUN
X μS/cm

- 5 Remove any tag indicating storage with disinfectant.

If detecting residuals

If residuals are detected after starting the WRO 300 by pressing  for three seconds, repeat the residual test every five minutes until no residuals can be detected.

3.1.4 Central Chem (Not available in the USA)

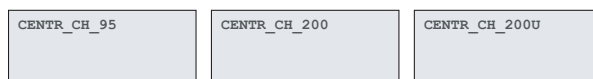
The Central Chem disinfects the WRO 300 and also sends disinfectant into the dialysis machine to enable Chemical disinfection of the inlet parts of this device.

For more detailed information on how to perform Central Chem together with Gambro Dialysis machines, please refer to *separate instructions for Central Chem*.

Procedure

Follow the *“Chemical disinfection procedure” on page 3:3*, but start at *step 2*.

In *step 7* select the preferred Central Chem protocol in the **CHEM SELECT** menu.



3.2 Cleaning

3.2.1 Acid cleaning

Acid cleaning should be performed when

- Rejection rate has decreased by more than 5 % from initial value.
- Sufficient output flow is not obtained.
- As preventive measure when a softener is not used.
- As a method to remove transitional metals from the RO membrane prior to Chemical disinfection.

Dissolve 6 table spoons of citric acid in 200 ml of product water from the WRO 300. This will result in a 2 % solution in the WRO 300.

For procedure refer to *“Cleaning procedure” on page 3:9*.

NOTE

Other acid membrane cleaner may be used as an alternative cleaner for the WRO 300. Follow *manufacturer's guidelines* to prepare the cleaning solution.

3.2.2 Alkaline cleaning

Alkaline cleaning should be performed when

- Product performance is affected and cleaning with citric acid does not improve performance.
- Organic fouling is suspected.

For procedure refer to *“Cleaning procedure” on page 3:9.*

3.2.3 Cleaning procedure

- 1 Start the WRO 300 in RUN. Wait until the product water conductivity stabilizes and record the value.
- 2 Follow the Chemical disinfection procedure, but instead select **ACID CLEAN** or **ALKALINE CLEAN** in the **CHEM SELECT** menu. Refer to *“Chemical disinfection procedure” on page 3:3.*

NOTE

The container can be empty before the intake phase is finished.

Test for residuals

The absence of cleaning chemical in the product water is confirmed if the conductivity of the product water corresponds to the previously recorded value and the pH is within 1.0 pH unit of the feed water.

3.3 Long time Storage

If the WRO 300 is to be taken out of operation for shorter periods of time Chemical disinfection can be used to maintain the microbiological condition. This should be performed on a weekly basis or according to the schedule based on a microbiological testing, refer to *“Chemical disinfection” on page 3:1*.

3.3.1 Preservation

If Chemical disinfection (according to *“Long time Storage” on page 3:10*) is not possible to perform when the WRO 300 is taken out of operation for an extended period of time a chemical preservation has to be done. For example if electrical and water connections are disconnected.

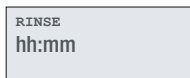
Contact Gambro for suitable chemical agents for the procedures.

Procedure

- 1 Follow the Chemical disinfection procedure (refer to *“Chemical disinfection” on page 3:1*) *step 1- step 11*, but instead select preferred preservation protocol in the **CHEM SELECT** menu.
- 2 After *step 11*: the buzzer will now sound and the indications  and  will flash. When  is pressed the display will show **404 RINSE REQUIRED**.
- 3 Switch off the WRO 300 on the mains power switch behind the detachable cover. Remove the angle connector from the Tank air vent connection and insert the plug provided with the WRO 300. Refer to *Installation Manual*.

3.3.2 Rinse after preservation

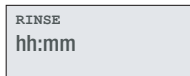
- 1 Remove the plug from Tank air vent connection and re-insert the angle connector.
- 2 Turn on the WRO 300 on the mains power switch
- 3 The indication  and  flashes and the display shows RINSE



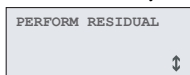
NOTE

If  is pressed the display will show **404 RINSE REQUIRED**.
Steady light on indication .

- 4 Press  for three seconds to initiate rinse. (Steady light of the indication .

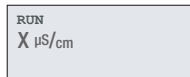


- 5 When the rinse program is completed the display will show **PERFORM RESIDUAL TEST**. The indication  will remain steady lit and the  will flash.



3.3.3 Residual test after preservation

- 1 Press  for three seconds to start WRO 300. Indication  will be steady lit and  will flash.
- 2 After a few minutes: Check and make sure that the product water conductivity corresponds to normal reading. Refer to *“Long time Storage” on page 3:10*. If not, let the WRO 300 continue to run until the conductivity value is normal.
- 3 Confirm by pressing  until light goes out (three seconds). Indication  remains steady lit.



- 4 Remove any tag indicating preservation.

3.3.4 Manual Rinse

In case a manual rinse has to be done follow the instruction below.

- 1 Insert the wand connector into the chemical intake port.
- 2 From *CHEM SELECT*, press down arrow button  to enter the list of available Cleaning and Chemical disinfection programs. Select for example Dialox.
- 3 Press  for three seconds to initiate the Chemical disinfection procedure.
- 4 Interrupt the procedure by disconnect the wand connector.
- 5 Press  for three seconds to start a rinse.

3.4 Exterior cleaning

Wipe the outside of the WRO 300 with a cloth moistened with ethanol (70 %) or isopropanol (60 %).

NOTE

Do not use iodine-based or tenside-containing disinfectants as these may crack or discolour the plastic materials.

4. Technical data

4.1 Performance and specification

		Values
Product water	Output flow	Min. 1.1 l/min at + 10 °C
	Product water loop	Maximum 20 meters (2 x 10m)
	Product water pressure	120 to 600 kPa (17 to 87 PSI) during RUN mode
	Quality	Depends on feed water quality. If potable water is used, the following rejection rates will be obtained:
		Total dissolved salts: > 96% (based on conductivity)
		Bacteria and endotoxins: > 99%

		Values
Water supply	Input	Min. 3.0 l/min required
	Feed Water Pressure	During operation: 100 to 300 kPa (15 to 44 PSI) (There is an inlet flow of water into the WRO)
		Maximum design pressure: 800 kPa (116 PSI)
		If the pressure exceeds 300 kPa (44 PSI), a pressure regulator, order no K23389 002, shall be installed.
	Feed Water Temperature	+ 5 to + 30 °C
	Feed Water Quality	Potable water should be used.
		Particle filtration shall be used (< 5µm)
		Carbon filtration shall be used if water is chlorinated, see limits below.
	Hardness	< 1° dH (20 ppm as CaCO ₃) 1 grain per gallon
	Iron	< 0.1 mg/l
	Manganese	< 0.1 mg/l
	Turbidity	< 1 JTU
	Total dissolved salts (TDS)	< 1500 mg/l
	Fouling index (silt density index)	< 5
	Chlorine (total)	< 0.1 mg/l
Drain	Drain output flow during RUN mode	1.2 ± 0.1 l/min
	Height difference between the drain hose outlet and the WRO 300 drain outlet.	Max. 1.0 m
	Drain (length of hose)	Max. 5 m
	Drain flow capacity	Min. 3.0 l/min required

		Values	
Tankair vent	This outlet connects the tank to the atmosphere. This outlet must be connected with an air gap. If a hose is connected to the Tank air vent the hose shall have a constant slope. NOTE Air will move back and forth in this line because of the variation of the level in the break tank.		
Fluid Connections	Designed for flexible, reinforced tubing 8 mm x 2.5 mm (5/16" I.D.)		
Conductivity measurement (Temperature compensated)	Product water conductivity	Operating range	1 - 500 µS/cm
		Accuracy	± 10 % or ± 10 µS/cm
	Feed water conductivity	Operating range	10 - 2000 µS/cm
		Accuracy	± 10% or ± 10 µS/cm
Temperature measurement	Operating range	0 - 105 °C	
	Accuracy	± 4 °C	
Reverse osmosis module	Membrane material	Polyamide, thin film composite	
	Membrane configuration	Spiral wound	
	Membrane pH tolerance	2 to 11	

		Values	
Power supply	Mains Voltage	100 to 240 V AC (line voltage range 85 to 264 V AC)	
	Frequency	50 or 60 Hz (line frequency range 45 to 65 Hz)	
	Power Consumption Max	WRO 300	570 W 100 - 240 V AC
	Appliance inlet	According to IEC 60320; C14	
	Cable	Cable with protective earth Conductor length max. 3.5 m minimum 1 mm ² (220-240 V AC) AWG 16 at 100 and 115 V AC	
	Mains plug	Continental Europe - 230 V AC, mains plug CEE-7 / VII United Kingdom - 230 V AC, mains plug BS 1363/A Australia - 240 V AC, mains plug AS/ NZS 3112 North America type - 100 and 115 V AC, Hospital grade, earthed plug, type IEC 60 083; A5-15	
	Fuses	Not used	
	Earth Leakage Current	Max. 250 µA (220-240 V AC) Max. 145 µA (115 V AC) Max. 140 µA (100 V AC)	

		Values	
Connection of external equipment	External connector 8 pins REDEL, yellow.		
	Remote start (Not available in the USA)	Max voltage	5 V DC
		Max current	100 mA DC
	External alarm	Max voltage	24 V AC or DC
		Max current	100 mA AC or DC
	External connector 8 pins REDEL, black. RS-232		
	Logging interface	Max input voltage	±15 V DC
		High level min output voltage	5.0 V DC
		Low level max output voltage	5.0 V DC
		Max Current	±5 mA DC
	External connector 8 pins REDEL, gray.		
	External equipment	Accessory equipment connected to the WRO 300 should comply with relevant standards (e.g. IEC 60950 for data processing equipment). Furthermore, medical systems should comply with IEC 60601-1-1. One important issue to consider, when configuring a system, is that the resulting leakage current for the system does not exceed acceptable limits.	
Sound	Sound pressure level	Less than 52 dB(A) during normal operation	

4.2 Chemical disinfection and Cleaning

Approved disinfectants	The following disinfectants may be administered via the chemical intake - Dialox, undiluted (Not available for sale in the USA) - Minncare® Cold Sterilant or Renalin 100 - Other peracetic acid disinfectants, provided that they are approved by the manufacturer for disinfection of thin film composite reverse osmosis membranes made of modified polyamide. Follow the manufacturer's instructions for Use for the specific disinfectant.	
Cleaning agents	Citric acid	
Exterior cleaning	Ethanol (70%) or isopropanol (60%).	

4.3 Physical data

Measurements	Values
Weight	29 kg
Height	563 mm
Depth	Max 520 mm
	Footprint 380 mm
Width	Max 205 mm
	Footprint 185 mm
Internal fluid volume	Approximately 4.5 litres

4.3.1 Materials in contact with product water

Type	Material
Polymers	PA (Polyamide) PEEK (Polyetheretherketone) PEI (Polyetherimide) PE (Polyethylene) POM (Polyoxymethylene) PP (Polypropylene) PPS (Polyphenylsulphite) PVC (Polyvinylchloride) PVDF (Polyvinylidenfluorid)
Rubber	EPDM (Ethylene-Propylene) NBR (Nitrile) Q (Silicone)
Metals	Titanium Stainless steel
Others	Ceramic

4.4 Environmental data

		Value
Operation	Ambient Temperature range	+ 10 to + 40 °C
	Relative Humidity range	30 to 85 % RH
	Air Pressure range	700 to 1060 hPa

		Value
Transportation and storage	Ambient Temperature range	- 10 to + 40 °C
	Relative Humidity range	10 to 95% RH
	Air Pressure range	500 to 1060 hPa
	For transportation and storage below the freezing point, the WRO 300 must be filled with a preservation solution of 3 % formalin (37 %), at least 9% glycerol and the remaining parts RO-water.	
	During transportation and storage the equipment must be kept in its original packing. If transportation or storage time is more than 15 weeks, the environmental data relating to the operation must be followed.	
	If condensation occurs when moving the equipment between locations with different temperatures and high relative humidity (e.g. outdoor and indoor locations), the inside of the equipment must be allowed to dry before switching on the equipment.	

4.4.1 Electromagnetic environment

		Value	
TheWRO 300 is intended for use in the electromagnetic environment specified below. The customer or the user of the WRO 300 should assure that it is used in such an environment.			
Emissions test	Compliance	Electromagnetic environment - guidance	
RF emissions CISPR 11	Group1	The WRO 300 uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment	
RF emissions CISPR 11	Class B	The WRO 300 is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes	
Harmonic emissions IEC 61000-3-2	Class A (Not applicable for 115V version)		
Voltage fluctuations/flicker emissions IEC 61000-3-3	Complies (Not applicable for 115 V version)		
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment-guidance
Electrostatic discharge (ESD)IEC 61000-4-2	±6 kV contact ±8 kV Air	±6 kV contact ±8 kV Air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Surge IEC 61000-4-5	±1kV for differential mode ±2kV for common mode	1kV for differential mode ±2kV for common mode	Mains power quality should be that of a typical commercial or hospital environment

		Value	
Voltage dips, short interruptions and voltage variations on power supply input lines. IEC 61000-4-11	<5% U_T ^a (>95% dip in U_T) for 0.5 cycle	<5% U_T (>95% dip in U_T) for 0.5 cycle	Mains power quality should be that of a typical commercial or hospital environment.
	40% U_T (60% dip in U_T) for 5 cycles	40% U_T (60% dip in U_T) for 5 cycles	
	70% U_T (30% dip in U_T) for 25 cycles	70% U_T (30% dip in U_T) for 25 cycles	
	<5% U_T (>95% dip in U_T) for 5 sec	<5% U_T (>95% dip in U_T) for 5 sec	
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment-guidance^b
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 V	Portable and mobile RF communications equipment should be used no closer to any part of the WRO 300, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance $d = 1.2 \sqrt{P}$
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 3.0 GHz	3 V/m	$d = 1.2 \sqrt{P}$ 80 to 800 MHz ^c $d = 2.3 \sqrt{P}$ 800 MHz to 2.5 GHz

		Value	
Radiated RF mobile phones	-	30 V/m	where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey^d, should be less than the compliance level in each frequency range^e. Interference may occur in the vicinity of equipment marked with the following symbol: 
Recommended separation distances between portable and mobile RF communications equipment and the WRO 300			
The WRO 300 is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the WRO 300 can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the WRO 300 as recommended below, according to the maximum output power of the communications equipment.			
Rated maximum output power of transmitter W	Separation distance according to frequency of transmitter (m)		
	150 kHz to 80 MHz	80 MHz to 800 MHz ^f	800 MHz to 3 GHz
	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{7}{3} \right] \sqrt{P}$
0,01	0.11	0.11	0.23
0,1	0.37	0.37	0.74
1	1.2	1.2	2.3
10	3.7	3.7	7.4
100	12	12	23

		Value	
Rated maximum output power of mobile phone	-	-	$d = \left[\frac{7}{30} \right] \sqrt{P}$
2W GSM/3G	-	-	0.33
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			

- NOTE:** U_T is the AC mains voltage prior to application of the test level.
- NOTE:** These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.
- NOTE:** At 80 MHz and 800 MHz, the higher frequency range applies.
- Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the WRO 300 is used exceeds the applicable RF compliance level above, the WRO 300 should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the WRO 300.
- Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.
- NOTE:** At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

4.5 Safety

The machine complies with the following standards:
IEC 60601-1 General requirements for safety, Class I, type B, and IPX1.
IEC 60601-1-2 Electromagnetic compatibility.

5.Troubleshooting

5.1 Alarms and notifications

No.	Alarm and notifications	Definition	
100	GENERAL		
101	CONDUCTIVITY SENSOR FAILURE	WRO 300	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	The initial conductivity system check failed.
		Action	Check that product water conductivity is within normal range, if so continue as normal. But inform technical personnel.
102	HIGH LEVEL SENSOR FAILURE	WRO 300	WRO 300 stops, buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Failure of top level sensor in tank.
		Action	Call for technical personnel. ^a
103	- N INTERNAL ERROR ^b	WRO 300	WRO 300 stops, buzzer sounds,  lights up, alarm info text after confirmation.
		Reason	Control unit failure.
		Action	Restart unit, or call for technical personnel.

- a. If the cover is removed during service external light may disturb the function of the level sensors.
- b. The number (N) shown before the alarm text identifies which specific kind of alarm that has occurred. Reference list for those number refer to *Service Manual*.

No.	Alarm and notifications	Definition	
200	OPERATION		
201	CONDUCTIVITY NOTIFICATION	WRO 300	Entering Standby mode, buzzer sounds,  flashes, info text after confirmation.
		Reason	Limit for elevated conductivity has been exceeded during operation.
		Action	1. Press  to silence buzzer 2. Notify technical personnel.
202	CONDUCTIVITY ALARM	WRO 300	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Limit for high conductivity has been exceeded.
		Action	1. Press  to silence buzzer. 2. Read conductivity values and call for technical personnel.
203	CONDUCTIVITY STOP	WRO 300	WRO 300 stops, buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Maximum safety limit for Conductivity has been exceeded.
		Action	1. Press  to silence buzzer. 2. Press  until the WRO 300 restarts (three seconds). If problem persists discontinue dialysis. 3. Call for technical personnel.
205	LOW REJECTION RATE	WRO 300	Entering Standby mode, buzzer sounds,  flashes, info text after confirmation.
		Reason	Decreased membrane performance.
		Action	Check conductivity values, Notify technical personnel

No.	Alarm and notifications	Definition	
400	CHEMICAL / RINSE		
401	REMOVE WAND	WRO 300	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Wand connector not removed after completion of Chemical intake.
		Action	Remove wand from disinfectant intake port.
402	INSUFFICIENT WATER SUPPLY	WRO 300	WRO 300 stops, buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Low level sensor indicates low level in the tank during the filling phase.
		Action	1. Press  to silence buzzer. 2. Check the feed water supply. 3. Check connections and unit for leakage. 4. Call for technical personnel.
403	HIGH CONDUCTIVITY RINSE	WRO 300	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Elevated product water conductivity during final part of rinse phase.
		Action	Perform an extra rinse by pressing  until rinse starts. If problem persists call technical personnel.
404	RINSE REQUIRED	WRO 300	Buzzer sounds,  and  flashes, alarm info text after confirmation.
		Reason	Machine filled with disinfectant.
		Action	Perform rinse by pressing  until rinse starts.

No.	Alarm and notifications	Definition	
405	INCORRECT WATER LEVEL	WRO 300	WRO 300 stops, buzzer sounds,  and  flashes, alarm info text after confirmation.
		Reason	Low level sensor indicates low level in the tank at a stage when the level is supposed to be stable.
		Action	1. Press  to silence buzzer. 2. Check that the dialysis machine is disconnected 3. Check connections and unit for leakage. 4. Call for technical personnel. 5. Press  to restart

No.	Alarm and notifications	Definition	
500	PRESERVATION		
501	REMOVE WAND	WRO 300	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Wand connector not removed after completion of Chemical intake.
		Action	Remove wand from disinfectant intake port.

5.2 Boot loader

The **LOAD APPLICATION FAILED** alarm indicate that the software is corrupt. If the alarm occurs in the display call for technical personnel.



6.Check list

Serial no:.....

Check conductivity with separate instrument

		Feed water		Product water		Reject		
Date	Hours in operation	Conductivity $\mu\text{S/cm}$	Temp $^{\circ}\text{C}$	Conductivity $\mu\text{S/cm}$	Flow l/min	Flow l/min	Rejection rate %	Remarks

7.Symbol description

NOTE
All symbols in this list may not be represented on this product.

7.1 Symbols

Symbol	Description
	Alternating current
	Protective earth (ground)
	Warning, consult accompanying documents
	Off (power, disconnection from the mains)
	On (power, connection to the mains)
	Type B, applied part
	Handle with care
	This way up
	Keep dry

Symbol	Description
	Separate collection for electrical and electronic equipment
	Input /Output
	Year of manufacturing
	Heater power supply outlet
	Conforms to requirements in EC Council Directive 93/42/EEC concerning medical devices. 0086 identifies BSI, British Standards Institution, as Notified Body.
	CSA approved for Canada
	CSA approved for Canada and USA
	CSA-listed label manufacturer
	Reject water connection
	Feed water inlet connection
	Loop connections (Product water outlet and return)